

Q8 - Formula of cost function (for linear regression)

$$\text{Ans} \rightarrow J(\theta) = \frac{1}{2m} \sum_{i=1}^m (h_{\theta}(x_i) - y_i)^2$$

where m is no. of training example

$h_{\theta}(x_i)$ is hypothesis (predicted value).

y_i is true value

Q9 - All formulae used in descriptive statistics.

Ans: Mean; $\bar{x} = \frac{\sum_{i=1}^n x_i}{n}$

* Median;

Case 1: (Odd No.)

$$\text{Median} = x_{\left(\frac{n+1}{2}\right)}$$

Case 2: Even No. = $\frac{x_{\left(\frac{n}{2}\right)} + x_{\left(\frac{n}{2}+1\right)}}{2}$

* Mode: Most frequent occurring value.

* Variance: $\sigma^2 = \frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n-1}$

* Standard Deviation

$$\sigma = \sqrt{\sigma^2}$$

* Range = Max(x) - Min(x)

* Interquartile Range (IQR) = $Q_3 - Q_1$
where Q is 3rd quartile & Q_1 is first quartile.

Q10. All formulae used in Inferential Statistics

Ans - $z\text{-score} = \frac{(x - \mu)}{\sigma}$

where; x is sampled value, μ is population mean,
 σ is population s.d

$\text{Confidence Interval} = \bar{x} \pm z \frac{\sigma}{\sqrt{n}}$

where, $\bar{x}$ is sampled mean, z is z -value from z -table),
 σ is population s.d & n is sample size.

$t\text{-score} = \frac{(\bar{x} - \mu)}{(s/\sqrt{n})}$

where, $\bar{x}$ is sample mean, μ is population mean,
 s is sample s.d, n is sample size

$\text{Chi-Square test} : \chi^2 = \sum \frac{(O_i - E_i)^2}{E_i}$

where; O_i is observed frequency, E_i is expected frequency

$\text{ANOVA (Analysis of Variance)} :$

$$F = \frac{MSB}{MSW}$$

where, MSB is mean sum of squares b/w groups.
& MSW is mean sum of squares within groups.

Q5- Formula of Tanh kernel.

Ans:- Formula $\rightarrow k(x, y) = \tanh(ax \cdot y + c)$

where;
 $\tanh$ is hyperbolic tangent function, a is scalar parameter & c is an offset.

Q6- Formula of Gradient descent.

Ans $\rightarrow \theta := \theta - \alpha \frac{\partial J(\theta)}{\partial \theta}$

where; θ is parameter vector, α is learning rate & $J(\theta)$ is cost function.

Q7. All three formulae of Linear Regression Metrics

Ans:- Mean Squared Error (MSE):

$$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

* R-Squared (Coefficient of determination)

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2}$$

* Mean Absolute Error (MAE):

$$MAE = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|$$

* RMSE (Root Mean Square Error):

$$RMSE = \sqrt{MSE}$$

where;
 y_i is true value
 $\hat{y}_i$ is predicted val.
 $\bar{y}$ is mean of true value.
 n is no. of observations.

ML formulae used:

Q1- Formula of Sigmoid function along with Range.

Ans: Formula; $\sigma(x) = \frac{1}{1+e^{-x}}$

& Range is $(0,1)$

Q2- Formula of Linear kernel along with Range.

Ans: $k(x,y) = x \cdot y$

Range $\rightarrow$ Any real number, it depends on the input vectors.

Q3- Formula used in Naive Bayes Classifier.

Ans: $P(A|B) = \frac{P(B|A) \cdot P(A)}{P(B)}$

Here;

$P(A/B)$ is posterior probability of given class A.

$P(B/A)$ is likelihood of vector B given class A.

$P(A)$ is the prior probability of class A

$P(B)$ is the marginal probability of vector B.

Q4- Formula of Polynomial Kernel.

Ans: Formula $\rightarrow k(x,y) = (x \cdot y + c)^d$

where;

x & y are input vectors.

c is constant

d is degree of polynomial.